Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

September 30, 2025

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Effluent PFAS
Pace Project No.: 50412958

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on September 18, 2025. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Minneapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

cc:

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

CERTIFICATIONS

Project:	Effluent PFAS
Pace Project No.:	50412958

Pace Analytical Services, LLC - Minneapolis MN

1700 Elm Street SE, Minneapolis, MN 55414
Alaska Contaminated Sites Certification #: 17-009
Arizona Certification #: AZ0014
Arkansas WW Certification #: 88-0680
Colorado Certification #: MN00064
DoD Certification via A2LA #: 2926.01
Florida Certification #: E87605
Idaho Certification #: MN00064
Indiana Certification #: C-MN-01
ISO/IEC 17025 Certification via A2LA #: 2926.01
Kentucky DW Certification #: 90062
Louisiana DEQ Certification #: AI-03086
Maine Certification #: MN00064
Michigan Certification #: 9909
Minnesota Dept of Ag Approval: via MN 027-053-137
Mississippi Certification #: MN00064
Montana Certification #: CERT0092
Nevada Certification #: MN00064
New Jersey Certification #: MN002
North Carolina DW Certification #: 27700
North Dakota Certification via A2LA #: R-036
Ohio DW Certification #: 41244
Oklahoma Certification #: 9507
Oregon Secondary Certification #: MN200001
Puerto Rico Certification #: MN00064
Tennessee Certification #: TN02818
Utah Certification #: MN00064
Virginia Certification #: 460163
West Virginia DEP Certification #: 382
Wisconsin Certification #: 999407970
USDA Permit #: P330-19-00208

Alabama Certification #: 40770
Alaska DW Certification #: MN00064
Arkansas DW Certification #: MN00064
California Certification #: 2929
Connecticut Certification #: PH-0256
EPA Region 8 Tribal Water Systems+Wyoming DW Certification #: via MN 027-053-137
Georgia Certification #: 959
Illinois Certification #: 200011
Iowa Certification #: 368
Kansas Certification #: E-10167
Kentucky WW Certification #: 90062
Louisiana DW Certification #: MN00064
Maryland Certification #: 322
Minnesota Certification #: 027-053-137
Minnesota Petrofund Registration #: 1240
Missouri Certification #: 10100
Nebraska Certification #: NE-OS-18-06
New Hampshire Certification #: 2081
New York Certification #: 11647
North Carolina WW Certification #: 530
North Dakota Certification via MN #: R-036
Ohio VAP Certification (1700) #: CL101
Oregon Primary Certification #: MN300001
Pennsylvania Certification #: 68-00563
South Carolina Certification #:74003001
Texas Certification #: T104704192
Vermont Certification #: VT-027053137
Washington Certification #: C486
West Virginia DW Certification #: 9952 C
Wyoming UST Certification via A2LA #: 2926.01

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Grand Rapids, MI 49512
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SAMPLE SUMMARY

Project: Effluent PFAS
Pace Project No.: 50412958

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50412958001	**Effluent**	Water	09/17/25 11:29	09/18/25 10:30

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SAMPLE ANALYTE COUNT

Project: Effluent PFAS
Pace Project No.: 50412958

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50412958001	**Effluent**	EPA 1633	MJL	65	PASI-M

PASI-M = Pace Analytical Services - Minneapolis

Date: 09/30/2025 01:00 PM

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SUMMARY OF DETECTION

Project: Effluent PFAS
Pace Project No.: 50412958

Lab Sample ID / Method	Client Sample ID / Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50412958001	**Effluent**					
EPA 1633	NMeFOSAA	1.1J	ng/L	1.6	09/27/25 02:09	
EPA 1633	PFBS	6.3	ng/L	1.6	09/27/25 02:09	
EPA 1633	PFDA	0.50J	ng/L	1.6	09/27/25 02:09	
EPA 1633	PFHxA	6.8	ng/L	1.6	09/27/25 02:09	
EPA 1633	PFBA	3.4J	ng/L	6.2	09/27/25 02:09	
EPA 1633	PFMPA	1.0J	ng/L	3.1	09/27/25 02:09	
EPA 1633	PFOSA	0.33J	ng/L	1.6	09/27/25 02:09	1d
EPA 1633	PFPeA	2.0J	ng/L	3.1	09/27/25 02:09	
EPA 1633	PFHpA	0.59J	ng/L	1.6	09/27/25 02:09	
EPA 1633	PFHxS	0.67J	ng/L	1.6	09/27/25 02:09	2d
EPA 1633	PFNA	0.56J	ng/L	1.6	09/27/25 02:09	
EPA 1633	PFOS	1.5J	ng/L	1.6	09/27/25 02:09	
EPA 1633	PFOA	2.1	ng/L	1.6	09/27/25 02:09	

Date: 09/30/2025 01:00 PM

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ANALYTICAL RESULTS

Project: Effluent PFAS
Pace Project No.: 50412958

Sample: Effluent **Lab ID: 50412958001** Collected: 09/17/25 11:29 Received: 09/18/25 10:30 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
EPA 1633 Water	Analytical Method: EPA 1633 Preparation Method: EPA 1633 Pace Analytical Services - Minneapolis								
11Cl-PF3OUdS	**<1.3**	ng/L	6.2	1.3	1	09/25/25 09:50	09/27/25 02:09	763051-92-9	
3:3 FTCA	**<2.0**	ng/L	7.8	2.0	1	09/25/25 09:50	09/27/25 02:09	356-02-5	
4:2 FTS	**<1.5**	ng/L	6.2	1.5	1	09/25/25 09:50	09/27/25 02:09	757124-72-4	
5:3 FTCA	**<5.3**	ng/L	38.9	5.3	1	09/25/25 09:50	09/27/25 02:09	914637-49-3	
6:2 FTS	**<1.4**	ng/L	6.2	1.4	1	09/25/25 09:50	09/27/25 02:09	27619-97-2	
7:3 FTCA	**<5.0**	ng/L	38.9	5.0	1	09/25/25 09:50	09/27/25 02:09	812-70-4	
8:2 FTS	**<2.4**	ng/L	6.2	2.4	1	09/25/25 09:50	09/27/25 02:09	39108-34-4	
9Cl-PF3ONS	**<1.1**	ng/L	6.2	1.1	1	09/25/25 09:50	09/27/25 02:09	756426-58-1	
ADONA	**<0.75**	ng/L	6.2	0.75	1	09/25/25 09:50	09/27/25 02:09	919005-14-4	
HFPO-DA	**<0.93**	ng/L	6.2	0.93	1	09/25/25 09:50	09/27/25 02:09	13252-13-6	
NEtFOSAA	**<0.42**	ng/L	1.6	0.42	1	09/25/25 09:50	09/27/25 02:09	2991-50-6	
NEtFOSA	**<0.31**	ng/L	1.6	0.31	1	09/25/25 09:50	09/27/25 02:09	4151-50-2	
NEtFOSE	**<3.7**	ng/L	15.5	3.7	1	09/25/25 09:50	09/27/25 02:09	1691-99-2	
NFDHA	**<0.79**	ng/L	3.1	0.79	1	09/25/25 09:50	09/27/25 02:09	151772-58-6	
NMeFOSAA	**1.1J**	ng/L	1.6	0.47	1	09/25/25 09:50	09/27/25 02:09	2355-31-9	
NMeFOSA	**<0.51**	ng/L	1.6	0.51	1	09/25/25 09:50	09/27/25 02:09	31506-32-8	
NMeFOSE	**<2.8**	ng/L	15.5	2.8	1	09/25/25 09:50	09/27/25 02:09	24448-09-7	
PFBS	**6.3**	ng/L	1.6	0.29	1	09/25/25 09:50	09/27/25 02:09	375-73-5	
PFDA	**0.50J**	ng/L	1.6	0.35	1	09/25/25 09:50	09/27/25 02:09	335-76-2	
PFHxA	**6.8**	ng/L	1.6	0.30	1	09/25/25 09:50	09/27/25 02:09	307-24-4	
PFBA	**3.4J**	ng/L	6.2	1.2	1	09/25/25 09:50	09/27/25 02:09	375-22-4	
PFDS	**<0.46**	ng/L	1.6	0.46	1	09/25/25 09:50	09/27/25 02:09	335-77-3	
PFDoS	**<0.33**	ng/L	1.6	0.33	1	09/25/25 09:50	09/27/25 02:09	79780-39-5	
PFEESA	**<0.78**	ng/L	3.1	0.78	1	09/25/25 09:50	09/27/25 02:09	113507-82-7	
PFHpS	**<0.41**	ng/L	1.6	0.41	1	09/25/25 09:50	09/27/25 02:09	375-92-8	
PFMBA	**<0.45**	ng/L	3.1	0.45	1	09/25/25 09:50	09/27/25 02:09	863090-89-5	
PFMPA	**1.0J**	ng/L	3.1	0.58	1	09/25/25 09:50	09/27/25 02:09	377-73-1	
PFNS	**<0.43**	ng/L	1.6	0.43	1	09/25/25 09:50	09/27/25 02:09	68259-12-1	
PFOSA	**0.33J**	ng/L	1.6	0.30	1	09/25/25 09:50	09/27/25 02:09	754-91-6	1d
PFPeA	**2.0J**	ng/L	3.1	0.43	1	09/25/25 09:50	09/27/25 02:09	2706-90-3	
PFPeS	**<0.25**	ng/L	1.6	0.25	1	09/25/25 09:50	09/27/25 02:09	2706-91-4	
PFDoA	**<0.33**	ng/L	1.6	0.33	1	09/25/25 09:50	09/27/25 02:09	307-55-1	
PFHpA	**0.59J**	ng/L	1.6	0.35	1	09/25/25 09:50	09/27/25 02:09	375-85-9	
PFHxS	**0.67J**	ng/L	1.6	0.41	1	09/25/25 09:50	09/27/25 02:09	355-46-4	2d
PFNA	**0.56J**	ng/L	1.6	0.31	1	09/25/25 09:50	09/27/25 02:09	375-95-1	
PFOS	**1.5J**	ng/L	1.6	0.52	1	09/25/25 09:50	09/27/25 02:09	1763-23-1	
PFOA	**2.1**	ng/L	1.6	0.45	1	09/25/25 09:50	09/27/25 02:09	335-67-1	
PFTeDA	**<0.26**	ng/L	1.6	0.26	1	09/25/25 09:50	09/27/25 02:09	376-06-7	
PFTrDA	**<0.31**	ng/L	1.6	0.31	1	09/25/25 09:50	09/27/25 02:09	72629-94-8	
PFUnA	**<0.34**	ng/L	1.6	0.34	1	09/25/25 09:50	09/27/25 02:09	2058-94-8	
Surrogates									
13C2-PFDoA (S)	51	%.	10-130		1	09/25/25 09:50	09/27/25 02:09		
13C3HFPO-DA (S)	60	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
13C3-PFBS (S)	53	%.	40-135		1	09/25/25 09:50	09/27/25 02:09		

REPORT OF LABORATORY ANALYSIS

Date: 09/30/2025 01:00 PM

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ANALYTICAL RESULTS

Project: Effluent PFAS
Pace Project No.: 50412958

Sample: Effluent **Lab ID: 50412958001** Collected: 09/17/25 11:29 Received: 09/18/25 10:30 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
Surrogates									
13C3-PFHxS (S)	50	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
13C4-PFBA (S)	70	%.	5-130		1	09/25/25 09:50	09/27/25 02:09		
13C4-PFHpA (S)	56	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
13C5-PFHxA (S)	64	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
13C5-PFPeA (S)	67	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
13C6-PFDA (S)	50	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
13C8-PFOA (S)	57	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
13C8-PFOS (S)	47	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
13C8-PFOSA (S)	55	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
13C9-PFNA (S)	53	%.	40-130		1	09/25/25 09:50	09/27/25 02:09		
d3-MeFOSAA (S)	60	%.	40-170		1	09/25/25 09:50	09/27/25 02:09		
d3-NMeFOSA (S)	53	%.	10-130		1	09/25/25 09:50	09/27/25 02:09		
d5-EtFOSAA (S)	60	%.	25-135		1	09/25/25 09:50	09/27/25 02:09		
d5-NEtFOSA (S)	50	%.	10-130		1	09/25/25 09:50	09/27/25 02:09		
d7-NMeFOSE (S)	64	%.	10-130		1	09/25/25 09:50	09/27/25 02:09		
d9-NEtFOSE (S)	58	%.	10-130		1	09/25/25 09:50	09/27/25 02:09		
13C2-PFTA (S)	40	%.	10-130		1	09/25/25 09:50	09/27/25 02:09		
13C7-PFUdA (S)	53	%.	30-130		1	09/25/25 09:50	09/27/25 02:09		
13C24:2FTS (S)	95	%.	40-200		1	09/25/25 09:50	09/27/25 02:09		
13C26:2FTS (S)	155	%.	40-200		1	09/25/25 09:50	09/27/25 02:09		
13C28:2FTS (S)	88	%.	40-300		1	09/25/25 09:50	09/27/25 02:09		
13C3-PFPrA (S)	25	%.	5-130		1	09/25/25 09:50	09/27/25 02:09		

Pace Analytical Services, LLC
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QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50412958

QC Batch 1031430
QC Batch Method: EPA 1633
Analysis Method: EPA 1633
Analysis Description: EPA 1633 Water
Laboratory: Pace Analytical Services - Minneapolis

Associated Lab Samples: 50412958001

METHOD BLANK: 5372101 Matrix: Water

Associated Lab Samples: 50412958001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
11Cl-PF3OUdS	ng/L	<1.3	6.4	1.3	09/26/25 23:08	
3:3 FTCA	ng/L	<2.1	8.0	2.1	09/26/25 23:08	
4:2 FTS	ng/L	<1.5	6.4	1.5	09/26/25 23:08	
5:3 FTCA	ng/L	<5.4	40.0	5.4	09/26/25 23:08	
6:2 FTS	ng/L	<1.5	6.4	1.5	09/26/25 23:08	
7:3 FTCA	ng/L	<5.2	40.0	5.2	09/26/25 23:08	
8:2 FTS	ng/L	<2.4	6.4	2.4	09/26/25 23:08	
9Cl-PF3ONS	ng/L	<1.2	6.4	1.2	09/26/25 23:08	
ADONA	ng/L	<0.78	6.4	0.78	09/26/25 23:08	
HFPO-DA	ng/L	<0.96	6.4	0.96	09/26/25 23:08	
NEtFOSA	ng/L	<0.32	1.6	0.32	09/26/25 23:08	
NEtFOSAA	ng/L	<0.43	1.6	0.43	09/26/25 23:08	
NEtFOSE	ng/L	<3.8	16.0	3.8	09/26/25 23:08	
NFDHA	ng/L	<0.81	3.2	0.81	09/26/25 23:08	
NMeFOSA	ng/L	<0.53	1.6	0.53	09/26/25 23:08	
NMeFOSAA	ng/L	<0.48	1.6	0.48	09/26/25 23:08	
NMeFOSE	ng/L	<2.9	16.0	2.9	09/26/25 23:08	
PFBA	ng/L	<1.3	6.4	1.3	09/26/25 23:08	
PFBS	ng/L	<0.30	1.6	0.30	09/26/25 23:08	
PFDA	ng/L	<0.36	1.6	0.36	09/26/25 23:08	
PFDoA	ng/L	<0.34	1.6	0.34	09/26/25 23:08	
PFDoS	ng/L	<0.34	1.6	0.34	09/26/25 23:08	
PFDS	ng/L	<0.47	1.6	0.47	09/26/25 23:08	
PFEESA	ng/L	<0.80	3.2	0.80	09/26/25 23:08	
PFHpA	ng/L	<0.36	1.6	0.36	09/26/25 23:08	
PFHpS	ng/L	<0.42	1.6	0.42	09/26/25 23:08	
PFHxA	ng/L	<0.31	1.6	0.31	09/26/25 23:08	
PFHxS	ng/L	<0.43	1.6	0.43	09/26/25 23:08	
PFMBA	ng/L	<0.47	3.2	0.47	09/26/25 23:08	
PFMPA	ng/L	<0.60	3.2	0.60	09/26/25 23:08	
PFNA	ng/L	<0.32	1.6	0.32	09/26/25 23:08	
PFNS	ng/L	<0.44	1.6	0.44	09/26/25 23:08	
PFOA	ng/L	<0.46	1.6	0.46	09/26/25 23:08	
PFOS	ng/L	<0.53	1.6	0.53	09/26/25 23:08	
PFOSA	ng/L	<0.31	1.6	0.31	09/26/25 23:08	
PFPeA	ng/L	<0.44	3.2	0.44	09/26/25 23:08	
PFPeS	ng/L	<0.26	1.6	0.26	09/26/25 23:08	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50412958

METHOD BLANK: 5372101 Matrix: Water

Associated Lab Samples: 50412958001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
PFTeDA	ng/L	<0.27	1.6	0.27	09/26/25 23:08	
PFTrDA	ng/L	<0.32	1.6	0.32	09/26/25 23:08	
PFUnA	ng/L	<0.35	1.6	0.35	09/26/25 23:08	
13C2-PFDoA (S)	%.	38	10-130		09/26/25 23:08	
13C2-PFTA (S)	%.	45	10-130		09/26/25 23:08	
13C24:2FTS (S)	%.	55	40-200		09/26/25 23:08	
13C26:2FTS (S)	%.	55	40-200		09/26/25 23:08	
13C28:2FTS (S)	%.	75	40-300		09/26/25 23:08	
13C3-PFBS (S)	%.	37	40-135		09/26/25 23:08	ES0
13C3-PFHxS (S)	%.	35	40-130		09/26/25 23:08	ES0
13C3-PFPrA (S)	%.	42	5-130		09/26/25 23:08	
13C3HFPO-DA (S)	%.	40	40-130		09/26/25 23:08	
13C4-PFBA (S)	%.	42	5-130		09/26/25 23:08	
13C4-PFHpA (S)	%.	38	40-130		09/26/25 23:08	ES0
13C5-PFHxA (S)	%.	40	40-130		09/26/25 23:08	
13C5-PFPeA (S)	%.	40	40-130		09/26/25 23:08	
13C6-PFDA (S)	%.	34	40-130		09/26/25 23:08	ES0
13C7-PFUdA (S)	%.	36	30-130		09/26/25 23:08	
13C8-PFOA (S)	%.	40	40-130		09/26/25 23:08	
13C8-PFOS (S)	%.	32	40-130		09/26/25 23:08	ES0
13C8-PFOSA (S)	%.	37	40-130		09/26/25 23:08	ES0
13C9-PFNA (S)	%.	36	40-130		09/26/25 23:08	ES0
d3-MeFOSAA (S)	%.	44	40-170		09/26/25 23:08	ES7
d3-NMeFOSA (S)	%.	31	10-130		09/26/25 23:08	
d5-EtFOSAA (S)	%.	45	25-135		09/26/25 23:08	ES7
d5-NEtFOSA (S)	%.	31	10-130		09/26/25 23:08	
d7-NMeFOSE (S)	%.	39	10-130		09/26/25 23:08	ES7
d9-NEtFOSE (S)	%.	41	10-130		09/26/25 23:08	ES7

LABORATORY CONTROL SAMPLE: 5372102

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
11Cl-PF3OUdS	ng/L	150	152	101	55-160	
3:3 FTCA	ng/L	198	208	105	65-130	
4:2 FTS	ng/L	150	157	104	70-145	
5:3 FTCA	ng/L	992	1080	109	70-135	
6:2 FTS	ng/L	154	164	107	65-155	
7:3 FTCA	ng/L	992	1040	105	50-145	
8:2 FTS	ng/L	154	166	108	60-150	
9Cl-PF3ONS	ng/L	150	159	106	70-155	
ADONA	ng/L	150	169	113	65-145	
HFPO-DA	ng/L	160	162	101	70-140	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

Date: 09/30/2025 01:00 PM

Pace Analytical Services, LLC
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QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50412958

LABORATORY CONTROL SAMPLE: 5372102

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
NEtFOSA	ng/L	38.4	40.9	107	65-145	
NEtFOSAA	ng/L	38.4	39.5	103	70-145	
NEtFOSE	ng/L	384	400	104	70-135	
NFDHA	ng/L	80	75.0	94	50-150	
NMeFOSA	ng/L	38.4	40.9	106	60-150	
NMeFOSAA	ng/L	38.4	40.0	104	50-140	
NMeFOSE	ng/L	384	397	103	70-145	
PFBA	ng/L	160	161	100	70-140	
PFBS	ng/L	35.2	35.8	102	60-145	
PFDA	ng/L	38.4	39.5	103	70-140	
PFDoA	ng/L	38.4	40.8	106	70-140	
PFDoS	ng/L	38.4	32.7	85	50-145	
PFDS	ng/L	38.4	34.7	90	60-145	
PFEESA	ng/L	70.4	67.4	96	70-140	
PFHpA	ng/L	38.4	40.4	105	70-150	
PFHpS	ng/L	38.4	39.4	103	70-150	
PFHxA	ng/L	38.4	40.1	104	70-145	
PFHxS	ng/L	35.2	36.9	105	65-145	
PFMBA	ng/L	80	82.0	102	60-150	
PFMPA	ng/L	80	79.5	99	55-140	
PFNA	ng/L	38.4	40.1	104	70-150	
PFNS	ng/L	38.4	37.6	98	65-145	
PFOA	ng/L	38.4	39.3	102	70-150	
PFOS	ng/L	38.4	34.8	91	55-150	
PFOSA	ng/L	38.4	40.7	106	70-145	
PFPeA	ng/L	80	80.2	100	65-135	
PFPeS	ng/L	38.4	38.6	101	65-140	
PFTeDA	ng/L	38.4	42.1	110	60-140	
PFTrDA	ng/L	38.4	39.5	103	65-140	
PFUnA	ng/L	38.4	39.7	103	70-145	
13C2-PFDoA (S)	%.			70	10-130	
13C2-PFTA (S)	%.			64	10-130	
13C24:2FTS (S)	%.			95	40-200	
13C26:2FTS (S)	%.			96	40-200	
13C28:2FTS (S)	%.			101	40-300	
13C3-PFBS (S)	%.			74	40-135	
13C3-PFHxS (S)	%.			71	40-130	
13C3-PFPrA (S)	%.			80	5-130	
13C3HFPO-DA (S)	%.			79	40-130	
13C4-PFBA (S)	%.			81	5-130	
13C4-PFHpA (S)	%.			77	40-130	
13C5-PFHxA (S)	%.			79	40-130	
13C5-PFPeA (S)	%.			78	40-130	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

Date: 09/30/2025 01:00 PM

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50412958

LABORATORY CONTROL SAMPLE: 5372102

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
13C6-PFDA (S)	%.			73	40-130	
13C7-PFUdA (S)	%.			73	30-130	
13C8-PFOA (S)	%.			78	40-130	
13C8-PFOS (S)	%.			70	40-130	
13C8-PFOSA (S)	%.			72	40-130	
13C9-PFNA (S)	%.			73	40-130	
d3-MeFOSAA (S)	%.			86	40-170	ES7
d3-NMeFOSA (S)	%.			61	10-130	
d5-EtFOSAA (S)	%.			86	25-135	ES7
d5-NEtFOSA (S)	%.			61	10-130	
d7-NMeFOSE (S)	%.			74	10-130	ES7
d9-NEtFOSE (S)	%.			75	10-130	ES7

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

Date: 09/30/2025 01:00 PM

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALIFIERS

Project: Effluent PFAS
Pace Project No.: 50412958

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	The ion ratio is outside control limits
2d	The ion ratio is outside control limits.
ES0	Extracted Internal Standard recovery outside laboratory control limits.
ES7	The extracted internal standard recovery in an associated CCV is outside control limits. The recovery of the associated target compounds are within control limits.

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Effluent PFAS
Pace Project No.: 50412958

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50412958001	**Effluent**	EPA 1633	1031430	EPA 1633	1032287

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Pace® Location Requested (City/State):
Pace Analytical Grand Rapids
4171 40th Street SE, Grand Rapids, MI 49512

CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

WO# : 50412958

50412958

Company Name:	City of Cadillac
Street Address:	1121 Plett Rd, Cadillac, MI 49601
Contact/Report To:	Cindy Tomaszewski
Phone #:	231-775-2368
E-Mail:	ctomaszewski@cadillac-mi.net
Cc E-Mail:	
Customer Project #:	
Project Name:	Effluent PFAS
Invoice To:	Cindy Tomaszewski
Invoice E-Mail:	ctomaszewski@cadillac-mi.net
Site Collection Info/Facility ID (as applicable):	
Purchase Order # (if applicable):	
Quote #:	
Time Zone Collected:	[ ] AK [ ] PT [ ] MT [ ] CT [X] ET
County / State origin of sample(s):	Michigan
Data Deliverables:	[ ] Level II [ ] Level III [ ] Level IV [ ] EQUIS [ ] Other
Regulatory Program (DW, RCRA, etc.) as applicable:	NPDES
Reportable	[ ] Yes [ ] No
Rush (Pre-approval required):	[ ] Same Day [ ] 1 Day [ ] 2 Day [ ] 3 Day [ ] Other ______
DW PWSID # or WW Permit # as applicable:	
Date Results Requested:	
Field Filtered (if applicable):	[ ] Yes [X] No
Analysis:	

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Waste Water (WW), Product (P), Soil/Solid (SS), Oil (OL), Wipe (WP), Tissue (TS), Bioassay (B), Vapor (V), Surface Water (SW),Sediment (SED), Sludge (SL), Caulk (CK), Leachate (LL), Biosolid (BS), Other (OT)

Specify Container Size **: 2

Identify Container Preservative Type***: 1

Analysis Requested: 1633 PFAS - Pace MN

**Container Size: (1) 1L, (2) 500mL, (3) 250mL, (4) 125mL, (5) 100mL, (6) 40mL vial, (7) EnCore, (8) TerraCore, (9) 90mL, (10) Other

*** Preservative Types: (1) None, (2) HNO3, (3) H2SO4, (4) HCl, (5) NaOH, (6) Zn Acetate, (7) NaHSO4, (8) Sod. Thiosulfate, (9) Ascorbic Acid, (10) MeOH, (11) Other

Lab Use Only

Proj. Mgr:	Brian Hall
AcctNum / Client ID:	
Table #:	
Profile / Template:	7675
Prelog / Bottle Ord. ID:	EZ 3124087

Preservation non-conformance identified for sample.

Customer Sample ID	Matrix *	Comp / Grab	Composite Start Date	Composite Start Time	Collected or Composite End Date	Collected or Composite End Time	# Cont.	Res. Chlorine Results	Res. Chlorine Units	1633 PFAS - Pace MN										Sample Comment
Effluent	WW	G	—	—	9-17-25	1129	2	–	–	X										

Additional Instructions from Pace®:
1633 PFAS - 40 Compounds
Requires Reporting by 10th of Following Month

Collected By: (Printed Name) TAUSHA LACZKO
Signature: T. Laczko

Customer Remarks / Special Conditions / Possible Hazards:

# Coolers:	Thermometer ID:	Correction Factor (°C):	Obs. Temp. (°C)	Corrected Temp. (°C)	On Ice:

Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:
T. Laczko/city of Cadillac	9-17-25 1235	Ship VIA UPS Ground	
		Pace	9.18.25 1030

Tracking Number:

Delivered by: [ ] In- Person [ ] Courier [ ] FedEX [ ] UPS [ ] Other

Page: 1 of 1

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace® Terms and Conditions found at https://www.pacelabs.com/resource-library/resource/pace-terms-and-conditions/

ENV-FRM-CORQ-0019_v02_110123 ©

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Pace

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 9.18.25	Evaluated By: WDC	**WO#: 50412958** PM: BJH Due Date: 10/10/25 CLIENT: GR-Cadillac	
Client: City of Cadillac	PM: BJH		
Lab Notified of Rush or Short Holds: YES ~~NO~~ NA			

Project Received Via: FedEx UPS Client Pace Courier Other: ______				Comments:
Custody Seal Present and Intact:	YES	NO	~~NA~~	
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	~~NA~~	
Short Hold Present (≤ 48 Hours):	YES	~~NO~~		
Sample Received in Hold:	~~YES~~	NO		
Custody Signature Present:	~~YES~~	NO		
Collector Signature Present:	~~YES~~	NO		
Sample Collected Today and On Ice:	YES	NO	~~N/A~~	
IR Gun #: 350 ~~351~~ Therm #: 282 283	Temp. should be 0°C - 6°C (Initial/Corrected)			
Ice Type: WET ~~Bagged~~ / WET Loose BLUE NONE	1. Cooler Temp. Upon Receipt: 2.5/2.3 °C			
Ice Location: TOP ~~BOTTOM~~ MIDDLE ~~DISPERSED~~	2. Cooler Temp. Upon Receipt: ______ °C			
Temp Blank Received:	~~YES~~	NO		
Sample Label Matches COC (ID/Date/Time):	~~YES~~	NO		
Container Intact:	~~YES~~	NO		
Correct Container:	~~YES~~	NO		
Sufficient Volume:	~~YES~~	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with pink sticker on the lid if unacceptable. **pH Strip Lot #:** ______ *Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl*	YES	NO	~~N/A~~	
Residual Chlorine Absent: **Cl_2 Strip Lot #:** ______ *Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide*	YES	NO	~~N/A~~	
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES	NO	~~N/A~~	
Trip Blank Received: HCl MeOH Other: ______	YES	~~NO~~	ON HOLD	

Comments:	3. Cooler Temp. Upon Receipt: ______ °C
	4. Cooler Temp. Upon Receipt: ______ °C
	Non-Conformance Form Required: YES ~~NO~~

F-GR-C-007-rev.09, 11Aug2025

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